



Machine Learning I: practice part

Nelly Condori Fernandez

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Methodology

Problem based learning

Students work in **pairs**

The problems are grouped in 3 assignments

20% of the final grade

Project based learning

Students work in **groups** (4-6 people)

It consists of writing a report, implementing the solution of a classification/prediction problem, presentation of work

30% of the final grade

Content

Unit 0.-Installation of Julia, Jupyter

Unit 1.- Introduction (data load, data type)

Unit 2.- Multilayer Perceptron

Unit 3.- Overfitting

Unit 4.- Metrics for binary classification
and multiclass classification

Unit 5.- Cross-validation

Unit 6.- The Scikit-Learn library

Unit 7.- Ensemble models

Unit 8.- Dimension reduction

Group assignment evaluation:

Item	Weight	Score
Report	50 %	1.5
Code	30%	0.9
Oral presentation	20%	0.6
Total		3 points

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The prediction problem must be solved by 4 different approaches:

- Neural Networks,
- SVM,
- Decision trees and
- kNN

Unit 5

Unit 6

One ensemble technique should be applied in each approach

Unit 7

Performance comparative evaluation
(Confusion matrices and metrics)

Unit 4

Dimensionality reduction

Unit 8

Resources

- [Find Open Datasets and Machine Learning Projects | Kaggle](#)
- Report template : [ACM Primary Article Template](#)
 - Editor: [Overleaf, Online LaTeX Editor](#) (Recommended)
- [How to write an academic report: Academic Phrasebank | The University of Manchester](#)
- Code editor: [Getting Started · Julia in VS Code \(julia-vscode.org\)](#)
- Git repository: [Source Control with Git in Visual Studio Code](#)